THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 2: **Polynomials** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a quadratic polynomial? 1

(a) $x^3 + 1$ **(b)** $2x^2 - \sqrt{3}x + 1$ **(c)** $x + \frac{1}{x}$ **(d)** $\sqrt{x} + 2$

2. The graph of $y = p(x)$ touches the x-axis at exactly one point and does not cross it. The number of zeroes of $p(x)$ is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

3. The zeroes of $x^2 - 9$ are: 1

(a) $3, 3$ **(b)** $3, -3$ **(c)** $9, -9$ **(d)** $0, 9$

4. The sum of the zeroes of $2x^2 - 8x + 6$ is: 1

(a) −4 **(b)** 4 **(c)** 3 **(d)** −3

5. If α, β are the zeroes of $x^2 - 7x + 12$, then $\alpha\beta$ is: 1

(a) 7 **(b)** −7 **(c)** 12 **(d)** −12

6. A student states: "The zeroes of $x^2 + 7x + 10$ are 2 and 5." The correct response is: 1

(a) Correct **(b)** Incorrect; zeroes are -2 and -5

(c) Only 2 is correct **(d)** Only 5 is correct

7. If -1 is a zero of $x^2 + kx - 4$, then k equals: 1

(a) 3 **(b)** −3 **(c)** 4 **(d)** −4

8. Two students form a polynomial with zeroes 2 and -3. Student A writes $x^2 + x - 6$; Student B writes $x^2 - x - 6$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

9. The maximum number of zeroes a cubic polynomial can have is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

10. If one zero of $ax^2 + bx + c$ $(a \neq 0)$ is zero, then: 1

(a) $a = 0$ **(b)** $b = 0$ **(c)** $c = 0$ **(d)** $b = c$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the zeroes of $4s^2 - 4s + 1$ and verify the relationship between the zeroes and the coefficients. 2

12. Find a quadratic polynomial whose sum and product of zeroes are $\sqrt{2}$ and $\frac{1}{3}$ respectively. 2

13. If α and β are the zeroes of $x^2 - 5x + 6$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. 2

14. A student claims that $p(x) = x^2 - 4x + 4$ has two distinct zeroes. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the zeroes of $x^2 - 3$ and verify the relationship between the zeroes and the coefficients. 3

16. If α and β are the zeroes of $x^2 - x - 2$, find a quadratic polynomial whose zeroes are $2\alpha + 1$ and $2\beta + 1$. 3

17. If the zeroes of $x^2 + (a+1)x + b$ are 2 and -3, find the values of a and b. 3

OR

If the sum of the zeroes of $kx^2 + 2x + 3k$ is equal to their product, find k. Also write the polynomial. 3

18. Find a cubic polynomial with the sum, sum of the products of its zeroes taken two at a time, and the product of its zeroes as $2, -7, -14$ respectively. 3

OR

Verify that $5, -2$ and $\frac{1}{3}$ are the zeroes of $3x^3 - 10x^2 - 27x + 10$. Also verify that the sum of the zeroes $= -\frac{b}{a}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Find the zeroes of $p(x) = 2x^2 - x - 6$ and verify the relationship between the zeroes and the coefficients. 5
(ii) Find a quadratic polynomial whose zeroes are double the zeroes of $p(x)$.

OR

Draw the graph of $y = x^2 - 2x - 3$ for $-2 \le x \le 4$. Find the zeroes from the graph and verify the relationship between the zeroes and the coefficients. 5

20. **Case study:** A garden arch is in the shape of a parabola given by $y = -x^2 + 6x - 5$ (in metres), where $y = 0$ is ground level. 5
(i) Find the points where the arch meets the ground.
(ii) Find the width of the arch at the ground.
(iii) Verify the sum and product of the zeroes.
(iv) Find the height of the arch at $x = 3$.

OR

If α and β are the zeroes of $p(x) = x^2 - 4x + 3$, find a quadratic polynomial whose zeroes are α^2 and β^2. Also find the value of $\alpha^3 + \beta^3$. 5

END OF QUESTION PAPER — **Total: 40 Marks**